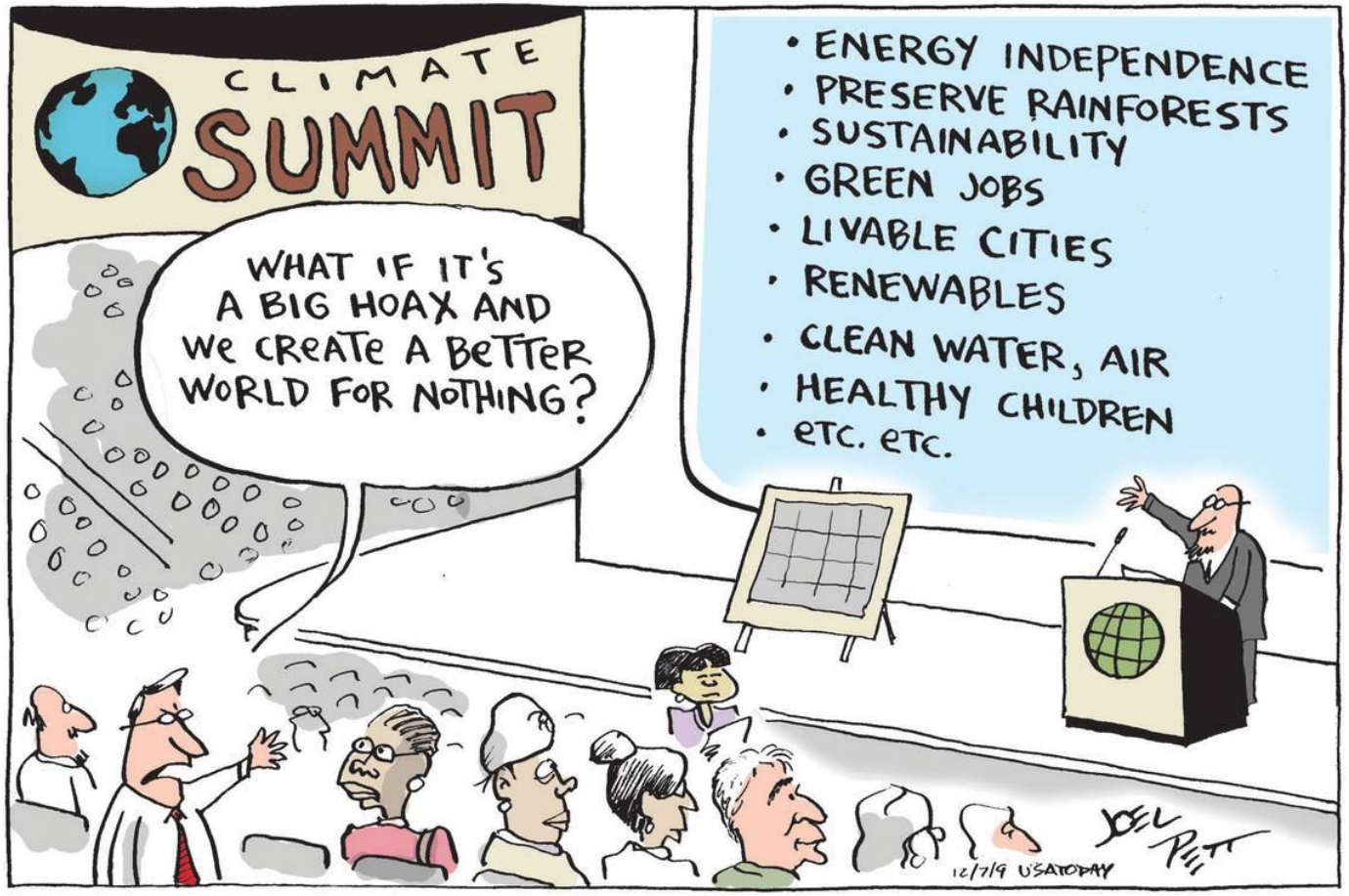
Can we put a price on climate impacts? *Social cost of carbon*

ECON-4210
Economics of Climate Change
Class 13



Prof. Casey Wichman

Georgia Institute of Technology

Today's class

1. Questions?
2. Attendance
3. Ghost emissions

   --> will keep open until Tuesday night… **Mayors, submit your decisions!!!**
4. Housekeeping
   1. Problem set 2 is due **Friday, 10/16, 11:59 PM.**
   2. Exam 2 is Monday, 10/19 (classes 9–13). Review on Wednesday, 10/14.
5. **Social Cost of Carbon**
6. For next class…

Where are we?

ECON-4210 so far…

- Basic climate science
- Climate change as an economic problem
- Benefit-cost analysis
- Discounting
- Damage functions and weather-response functions
- Damages to GDP and agriculture, country-level SCCs
- Heat-related mortality and the value of a statistical life

← Key components to understand the SCC

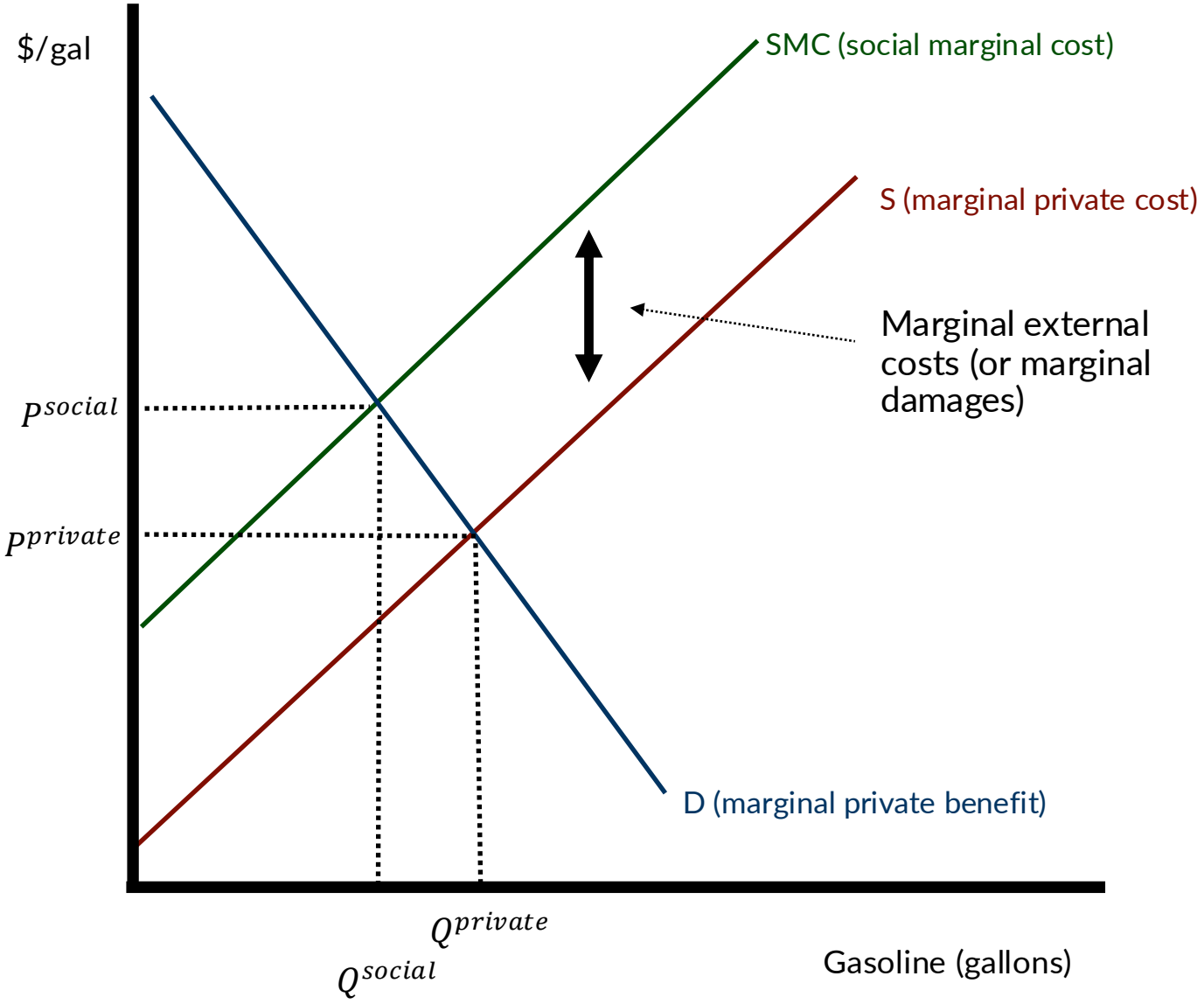
- **SOCIAL COST OF CARBON**

- Exam 2 (Monday, 10/19)

The **Social Cost of Carbon (SCC)*** is an estimate of the economic *damages* that would result from emitting *one additional ton of* CO_2 into the atmosphere

* For our purposes, the “Social Cost of Carbon” = the “Social Cost of Greenhouse Gases” = the “Social Cost of Carbon-Dioxide”

Marginal external costs



- Why do we care about climate damages?
- Marginal external costs (or marginal damages) are the (vertical) difference between the MC and SMC curves.
- Imagine the SCC is $200/ton.
  - There are 8887 grams (or 0.008887 tons) of CO2 in each gallon of gasoline.
  - $200/ton * 0.008887 CO2/gallon
  - → **$1.78/gallon in marginal external climate damages**

https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle

Some nomenclature…

- **Climate impacts:** the effect of climate change on some physical outcome (e.g., sea-level rise)
- **Climate damages:** the *monetized* economic effects of climate impacts.
- **Integrated assessment models (IAMs):** Computer models that incorporate key features of physical and economic realities to produce insights for decision-making. In our case, IAMs combine scientific and socioeconomic aspects of climate change for the purpose of assessing impacts and policies.

History of the SCC

2017:
Trump EO-13783 disbands IWG. Citing OMB's Circular A-4, directs agencies to use a 7% discount rate and ignore climate damages outside the US. *($1/ton, 7% discount rate)*

2008:
Federal appeals court ruled that federal agencies must monetize climate-related benefits in RIAs

2013 - 2016:
Updates to modeling process leading to revised SCC estimates *($42/ton in 2020 at 3% discount rate in $2007)*

2021:
In EO-13990, Biden reverses Trump EO, gets the IWG gang back together, calls for immediate reinstatement of SCC + more comprehensive update. ($51/ton in 2020 at 3% discount rate in $2020)

...

←1990s & 2000s:
Independent research / model development to estimate the SCC

2017:
National Academies study providing guidance on how the SCC can be updated to reflect the best available science.
(Prof. Wichman worked on this!)
https://www.nationalacademies.org/read/24651

2023:
Comprehensive update to SCC estimation process.
New central estimate of:
$190/ton at 2% discount rate

2010:
Interagency Working Group (IWG) created to harmonize estimates of the SCC across agencies

First systematic SCC estimates provided by the IWG *($26/ton in 2020 at 3% discount rate in $2007)*

Other notable things about the SCC:

- Some states also use it for energy policies.
- The Canadian government adopted its methodology.
- Mexican government is considering adopting the US SCC.
- Often used as a starting point for the level of a carbon tax.
- Influential for international negotiations

Georgia Institute of Technology

History of the SCC



An official website of the United States government Here's how you know

EPA United States Environmental Protection Agency

Search EPA.gov

Environmental Topics | Laws & Regulations | Report a Violation | About EPA

Home / News Releases

Related Links

Headquarters | Office of the Administrator (AO) | Office of Policy (OP)

Read other EPA News Releases about Administrator

Contact Us about News Releases

EPA Announces Action to Address Costly Obama, Biden "Climate" Measurements (Social Cost of Carbon)

March 12, 2025

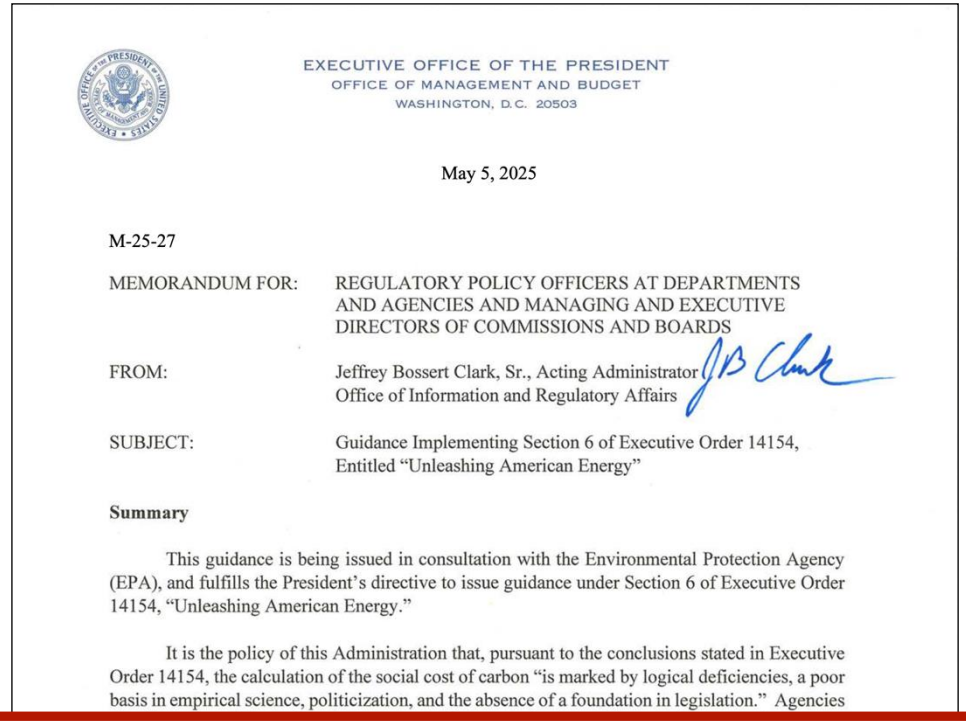
Contact Information
EPA Press Office (press@epa.gov)

WASHINGTON – U.S. Environmental Protection Agency (EPA) A
revisiting the Biden-Harris Administration's "social cost of carb

"The Biden-Harris Administration's so-called 'social cost of carb
way that imposed major costs. To Power the Great American Co
back the U.S.," ***said EPA Administrator Zeldin.***

The "Unleashing American Energy" Executive Order directs EPA to "issue guidance to address these harmful and detrimental inadequacies, including consideration of eliminating the 'social cost of carbon' calculation from any Federal permitting or regulatory decision" within 60 days. The Executive Order states, "[t]he calculation of the 'social cost of carbon' is marked by logical deficiencies, a poor basis in empirical science, politicization, and the absence of a foundation in legislation. Its abuse arbitrarily slows regulatory decisions and, by rendering the United States economy internationally uncompetitive, encourages a greater human impact on the environment by affording less efficient foreign energy producers a greater share of the global energy and natural resource market."

This was announced in conjunction with a number of historic actions to advance President Trump's Day One executive orders and Power the Great American Comeback. Combined, these announcements represent the greatest and most consequential day of deregulation in the history of the United States. While accomplishing EPA's core mission of protecting the environment, the agency is committed to fulfilling President Trump's promise to unleash American energy, lower costs for Americans, revitalize the American auto industry, restore the rule of law, and give power back to states to make their own decisions.



EXECUTIVE OFFICE OF THE PRESIDENT
OFFICE OF MANAGEMENT AND BUDGET
WASHINGTON, D.C. 20503

May 5, 2025

M-25-27

MEMORANDUM FOR: REGULATORY POLICY OFFICERS AT DEPARTMENTS AND AGENCIES AND MANAGING AND EXECUTIVE DIRECTORS OF COMMISSIONS AND BOARDS

FROM: Jeffrey Bossert Clark, Sr., Acting Administrator
Office of Information and Regulatory Affairs

SUBJECT: Guidance Implementing Section 6 of Executive Order 14154, Entitled "Unleashing American Energy"

Summary

This guidance is being issued in consultation with the Environmental Protection Agency (EPA), and fulfills the President's directive to issue guidance under Section 6 of Executive Order 14154, "Unleashing American Energy."

It is the policy of this Administration that, pursuant to the conclusions stated in Executive Order 14154, the calculation of the social cost of carbon "is marked by logical deficiencies, a poor basis in empirical science, politicization, and the absence of a foundation in legislation." Agencies

EO 14154 accordingly disbanded the Interagency Working Group on the Social Cost of Greenhouse Gases (IWG), which was established pursuant to EO 13990, and withdrew any guidance, instruction, recommendation, or document issued by the IWG. As such, it is no longer Federal government policy to maintain a uniform estimate of the monetized impacts of greenhouse gas emissions.

Sources:

- https://www.epa.gov/newsreleases/epa-announces-action-address-costly-obama-biden-climate-measurements-social-cost
- https://www.whitehouse.gov/wp-content/uploads/2025/02/M-25-27-Guidance-Implementing-Section-6-of-Executive-Order-14154-Entitled-Unleashing-American-Energy.pdf

Georgia Institute of Technology

Where things stand: fall 2026

- Jan 2025: EO 14154 disbands the IWG and withdraws the federal SCC estimates.
  - Agencies revert to the 2003 Circular A-4 (3% and 7% discount rates).
- May 2025: OMB tells agencies not to monetize climate damages unless a statute requires it.
- Feb 2026: EPA rescinds the 2009 **Endangerment Finding** and all vehicle GHG standards.
  - Rationale: Clean Air Act §202(a) does not cover global climate change.
  - EPA claims $1.3 trillion in savings, with no value on the added emissions.
- Sep 2026: EPA proposes the same rescission for power plants (§111).
- About 25 states and health/environmental groups have sued in the D.C. Circuit Court. Pending.

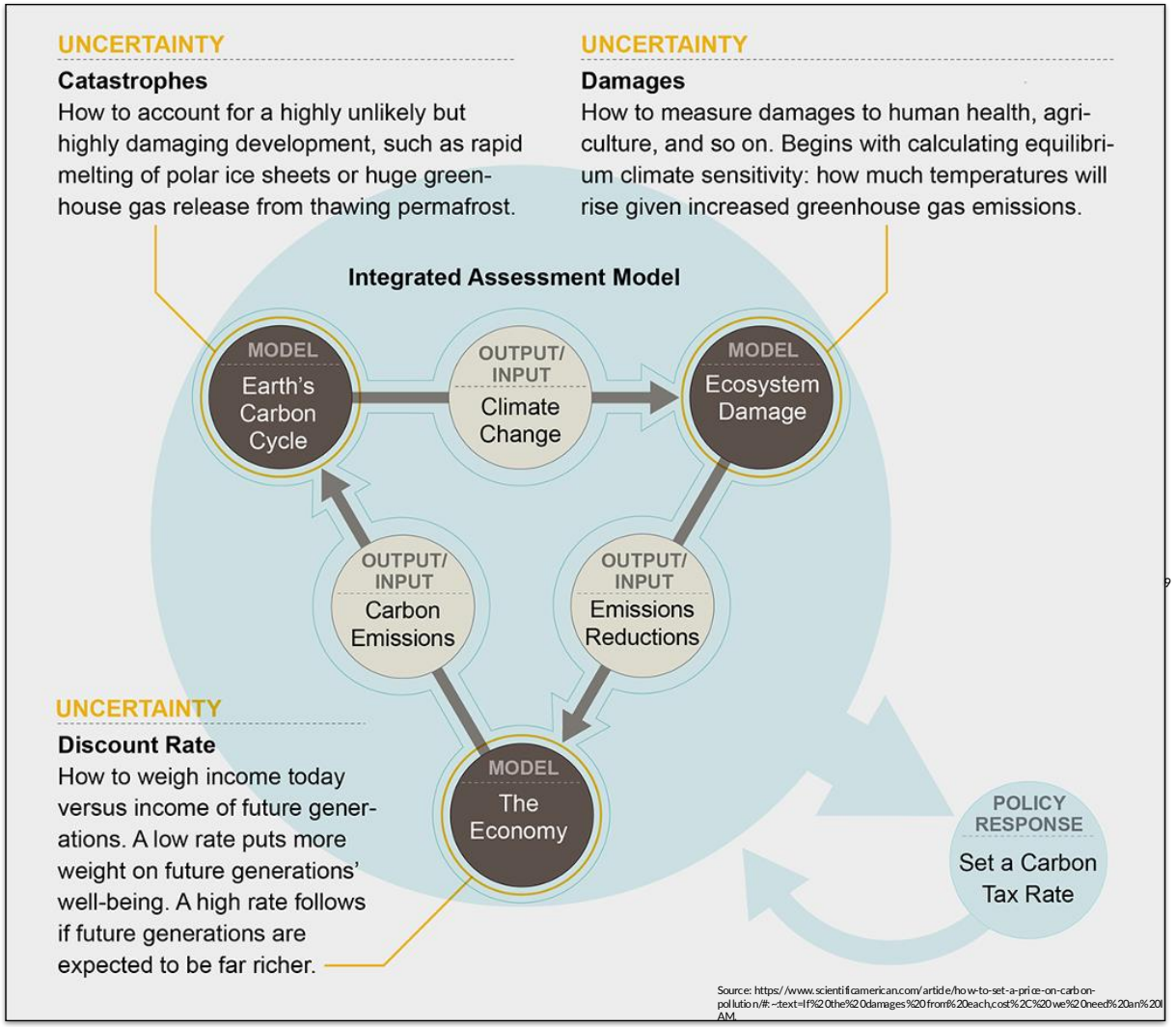
What does the endangerment finding have to do with the SCC?

Harvard SCC tracker: https://eelp.law.harvard.edu/tracker/the-social-cost-of-carbon/

Integrated assessment models (IAMs)

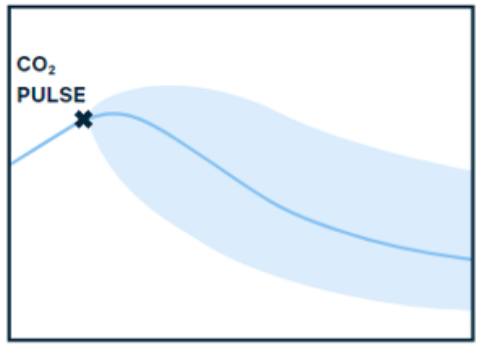
- IAMs applied to climate change
  - Represent key physical/economic features
  - Assess economic importance of climate change
  - Can be used to estimate the social cost of carbon
  - Allows for examining uncertainty in climate impacts
  - IAMS are *necessarily* a simplification of the real world

- There are many IAMs out there.
- Three notable IAMs have been used by the IWG to estimate the SCC historically.
  - DICE, FUND, and PAGE.
  - Each was developed by a single researcher (or in small teams) and there are many versions with different bells and whistles.

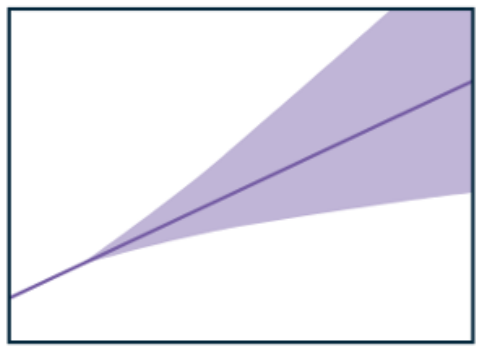


SOCIOECONOMIC MODULE

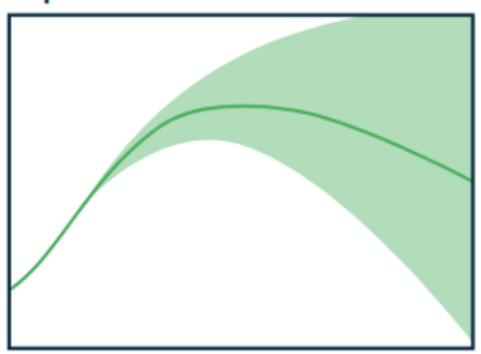
Emissions



Income

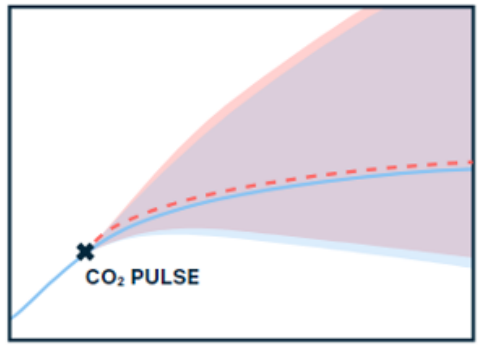


Population

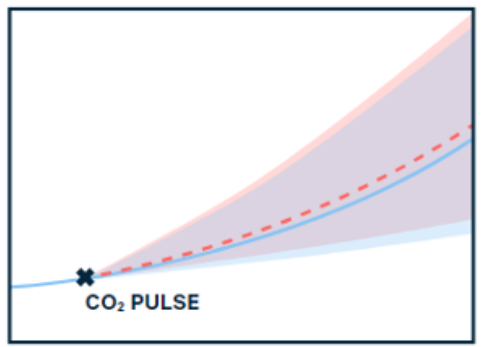


CLIMATE MODULE

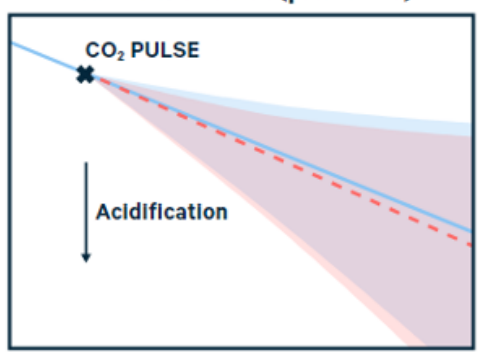
Temperature



Sea Level Rise

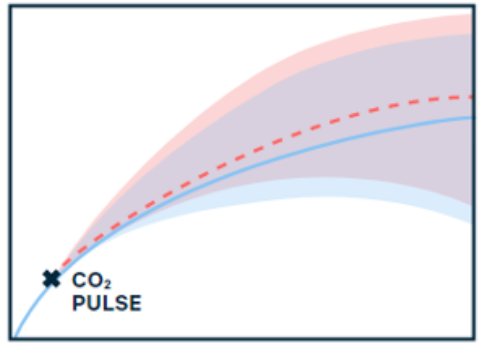


Ocean Acidification (pH Level)



DAMAGES MODULE

Global Climate Damages



Human Mortality Damages



Agriculture Sector Damages

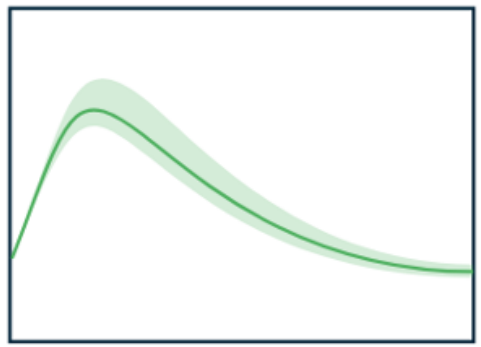


Other Damages

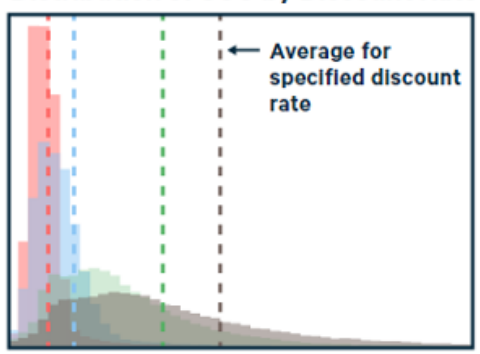


DISCOUNTING MODULE

Discounted Marginal Damage



Distribution of SCC by Discount Rate



SOCIAL COST OF CARBON

Georgia Institute of Technology

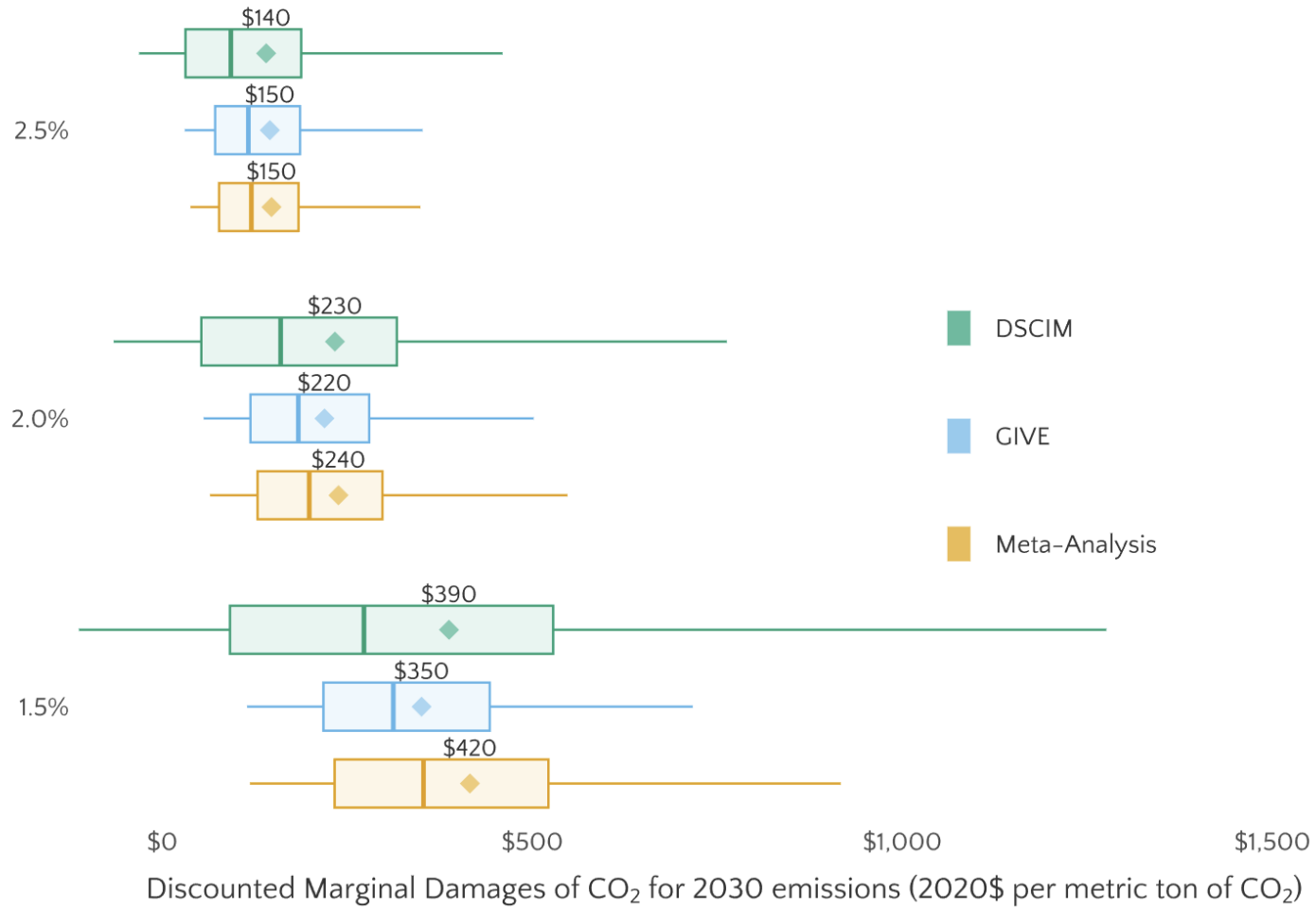
Damage functions in the SCC

- Earlier EPA SCC estimates (2010–2021) averaged three IAMs: DICE, FUND, and PAGE.
  - Each used a simple damage function of global mean temperature.
  - Calibrated to a small set of older studies.
  - Implied a 2–3% loss in global GDP by the end of the century.

- The 2023 EPA update used empirical damage functions instead.
  - Estimated from weather-response functions, the same approach as classes 10–12.
  - Built sector by sector: agriculture, mortality, energy, sea-level rise.
  - EPA averages three damage modules: GIVE (i.e., Rennert et al. (2022) -- we'll talk about this…), DSCIM, and a meta-analysis.

“Official” estimates of the SCC circa 2023

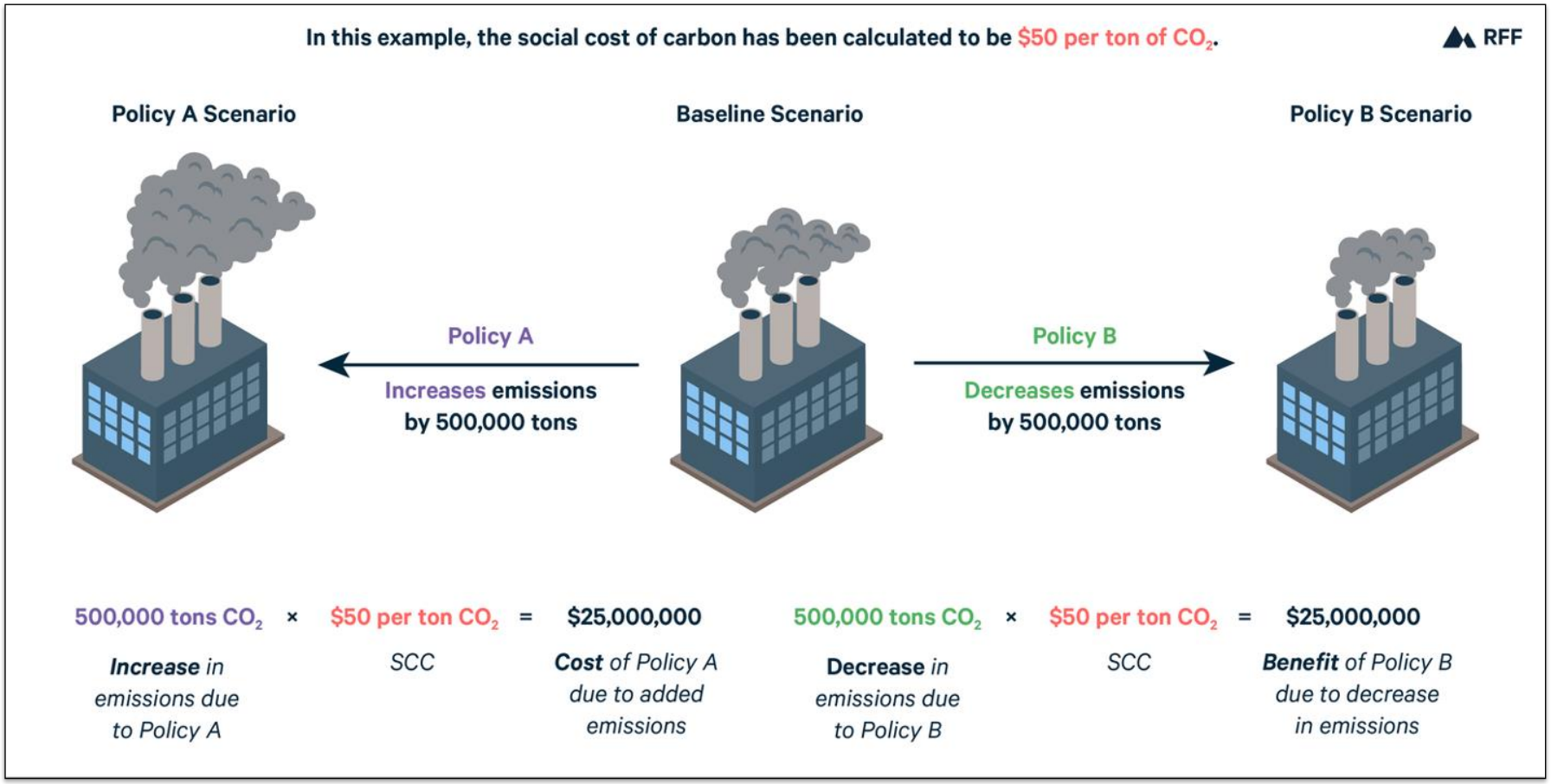
Figure 3.1.1: Distribution of the Discounted Marginal Damages per Metric Ton of Carbon Dioxide (CO_2) for 2030 Emissions, by Near-term Ramsey Discount Rate and Damage Module



Boxes span the inner quartile range (25th to 75th percentiles), whiskers extend to the 5th (left) and the 95th (right) percentiles. The vertical lines inside of the boxes mark the median of each distribution, and the points inside of the boxes and dollar estimates on top of the boxes mark the certainty-equivilant social cost of carbon (SC-CO_2)

Georgia Institute of Technology

How to use the SCC



- The SCC is the social cost of an additional ton of CO_2 emitted.
- If the SCC is $\$50$/ton and a policy *avoids* emitting x tons of CO_2, then the climate benefits of that policy are $\$50x$.
- Need to take into account *when* CO_2 emissions occur.

Domestic vs. global SCC

- One of the changes the Trump 1.0 admin made to the SCC was only counting damages that accrue within US borders.
  - Does this make sense?

Domestic vs. global SCC

- One of the changes the Trump 1.0 admin made to the SCC was only counting damages that accrue within US borders.
  - Does this make sense?

- On one hand, regulatory statutes in the US focus on domestic impacts only.
- On the other, GHG emissions are a *global* externality….
  - Climate impacts outside the US affect the US directly and indirectly
  - US citizens live abroad, thus might be affected by climate change
  - US has direct investments abroad, and we live in a global, interconnected economy
  - Climate change can destabilize regions, leading to migration and humanitarian concerns

- **2016 SCC at 3% → $51/ton. Domestic-only SCC → $7/ton. Discounting at 7% → $1/ton.**

2023 update to the SCC

- Rennert et al. build a new "modular" IAM to respond to National Academies' recommendations.
  - Open-source modeling framework to allow other researcher teams to easily add, e.g., a new damage function.
- Incorporates uncertainty via long-run probabilistic socioeconomic/emissions projections
- Decomposes damages regionally for 4 sectors.
- Builds in Ramsey-like discounting.
- New preliminary SCC estimate: **$185 per ton (2% discount rate)**

nature

Explore content · About the journal · Publish with us

nature › articles › article

Article | Published: 01 September 2022

Comprehensive Evidence Implies a Higher Social Cost of CO_2

Kevin Rennert, Frank Errickson, Brian C. Prest, Lisa Rennels, Richard G. Newell, William Pizer, Cora Kingdon, Jordan Wingenroth, Roger Cooke, Bryan Parthum, David Smith, Kevin Cromar, Delavane Diaz, Frances C. Moore, Ulrich K. Müller, Richard J. Plevin, Adrian E. Raftery, Hana Ševčíková, Hannah Sheets, James H. Stock, Tammy Tan, Mark Watson, Tony E. Wong & David Anthoff

Nature (2022) | Cite this article

4189 Accesses | **1552** Altmetric | Metrics

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

Abstract

The social cost of carbon dioxide (SC-CO_2) measures the monetized value of the damages to society caused by an incremental metric tonne of CO_2 emissions and is a key metric informing climate policy. Used by governments and other decision-makers in benefit-cost analysis for over a decade, SC-CO_2 estimates draw on climate science, economics, demography, and other disciplines. However, a 2017 report by the US National Academies of Sciences, Engineering, and Medicine[1] (NASEM) highlighted that current SC-CO_2 estimates no longer reflect the latest research. The report provided a series of recommendations for improving the scientific basis, transparency, and uncertainty characterization of SC-CO_2 estimates. Here we show that improved probabilistic socioeconomic projections, climate models, damage functions, and discounting methods that collectively reflect theoretically consistent valuation of risk, substantially increase estimates of the SC-CO_2. Our preferred mean SC-CO_2 estimate is $185 per tonne of CO_2 ($44-413/t-CO_2: 5-95% range, 2020 US dollars) at a near-term risk-free discount rate of 2 percent, a value 3.6-times higher than the US government's current value of $51/t-$CO_2$. Our estimates incorporate updated scientific understanding throughout all components of SC-CO_2 estimation in the new open-source GIVE model, in a manner fully responsive to the near-term NASEM recommendations. Our higher SC-CO_2 values, compared to estimates currently used in policy evaluation, substantially increase the estimated benefits of greenhouse gas mitigation and thereby increase the expected net benefits of more stringent climate policies.

https://www.nature.com/articles/s41586-022-05224-9

Rennert et al., *Nature* (2022)

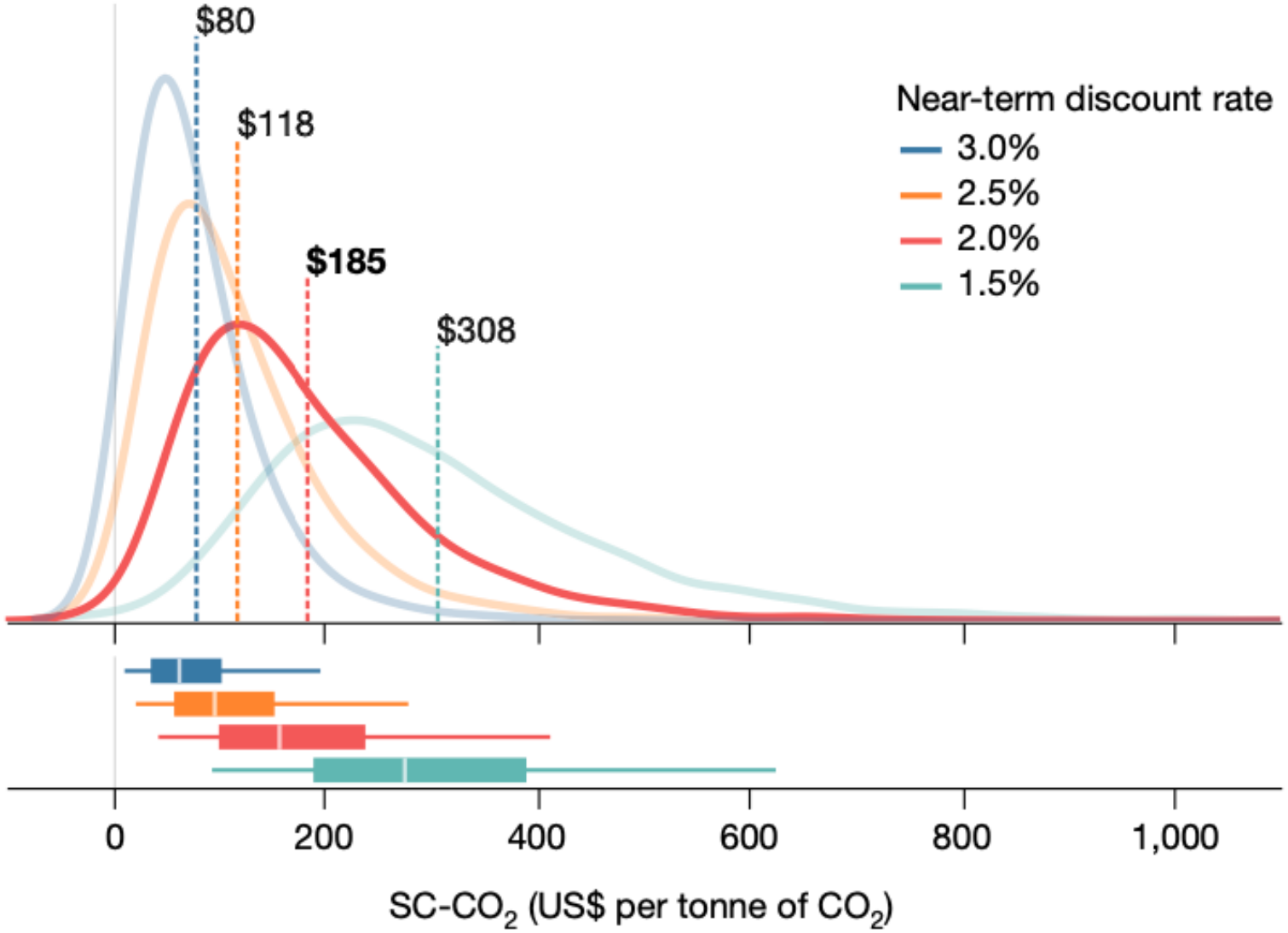


Fig. 2 | SC-CO_2 distributions vary with the choice of near-term discount rates. Distributions of the SC-CO_2 based on RFF-SP scenario samples,

Note: This paper adopts a version of Ramsey discounting ($r = \rho + \eta g$).

Their 2% discount rate implies:

$\rho = 0.2\%$ (pure rate of time preference)

$\eta = 1.24$ (inequality aversion)

Rennert et al., *Nature* (2022)

- Socioeconomic scenarios (coupled projections of country-level GDP, population, and emissions)
  - Statistical + structured expert elicitation for 300 years

- Global population peaks 2100-2150
- 1-2% long-run economic growth rate
- CO2 emissions decline over next 100 years (inclusive of climate policy)

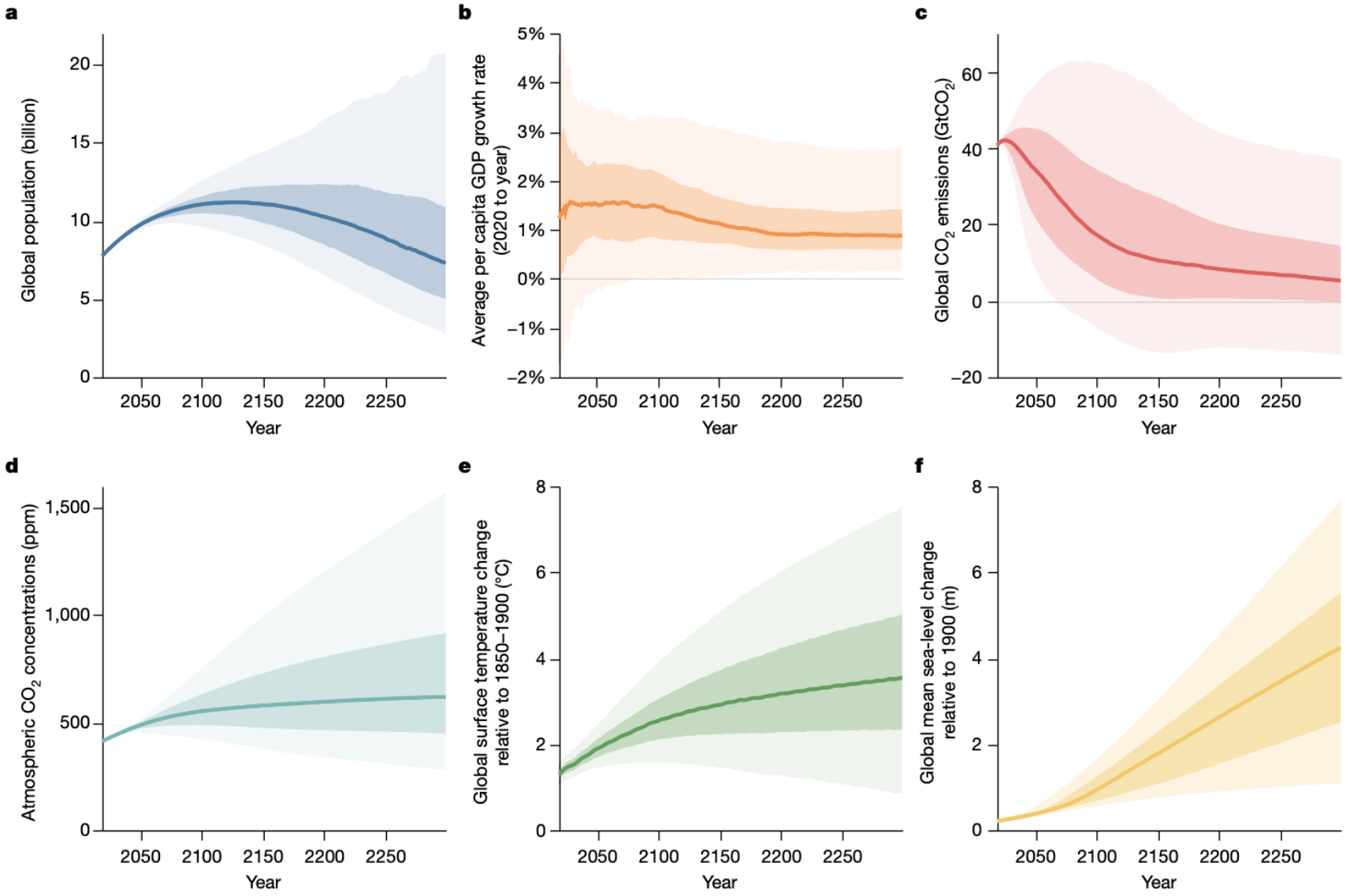


Fig. 1 | RFF-SP socioeconomic scenarios and the resulting climate system projections. a–c, Probabilistic socioeconomic projections for global population (**a**), per capita GDP growth rates (**b**), and carbon dioxide emission levels (**c**) from the RFF-SP scenarios. **d–f**, Corresponding climate system projections that account for parametric uncertainty in FaIR and BRICK for atmospheric carbon dioxide concentrations (**d**), global surface temperature changes relative to the 1850–1900 mean (**e**), and global mean sea-level changes relative to 1900 (**f**). In all panels, solid centre lines depict the median outcome, with darker shading spanning the 25%–75% quantile range and lighter shading spanning the 5%–95% quantile range.

Changes in SCC

- Comparison to existing IAMs
- DICE (used in previous SCC estimation) with same assumptions:
  - $44/ton
- GIVE with same damage function/discount rate as DICE:
  - $59/ton
- Adding in sectoral damages:
  - $80/ton
- GIVE with 2% discount rate:
  - $185/ton

Table 1 | Evolution of mean SC-CO_2 from DICE-2016R to this study

Row	Scenario	Mean SC-CO_2 ($ per tCO2)	Incremental change ($ per tCO2)	Share of total change (%)
a	DICE-2016R	44		
b	GIVE with DICE damage function, 3% near-term discount rate	59	15	11
c	GIVE with sectoral damages, 3% near-term discount rate	80	21	15
d	This study: GIVE with sectoral damages, 2% near-term discount rate	185	105	74

All SC-CO_2 values are expressed in 2020 US dollars per metric tonne of CO_2. Row a represents the SC-CO_2 using base DICE-2016R deterministic. The mean SC-CO_2 of $44 per tCO_2 is similar to the value previously estimated from IWG DICE-2010 of $46 per tCO_2 at a 3% discount rate, after converting to 2020 dollars[65]. Row b then retains the DICE-2016R damage function but otherwise deploys GIVE under discounting parameters of ρ=0.8%, η=1.57, which are consistent with a 3% near-term discount rate (see Methods section 'Discounting' for descriptions of ρ and η). Row c replaces the DICE-2016R damage function with our sectoral damage functions, and row d then uses our preferred discounting parameters from this study of ρ=0.2%, η=1.24, which are consistent with a 2% near-term discount rate. The final row represents the preferred mean value from this study.

Climate damages (in Rennert et al., 2022)

- Damage sectors include:
  - Agriculture (class 11)
  - Energy expenditures (class 10)
  - Mortality (heat deaths × VSL, class 12)
  - Sea-level rise (inclusive of adaptation)

- Mortality + agriculture = 94% of total
- What's missing?

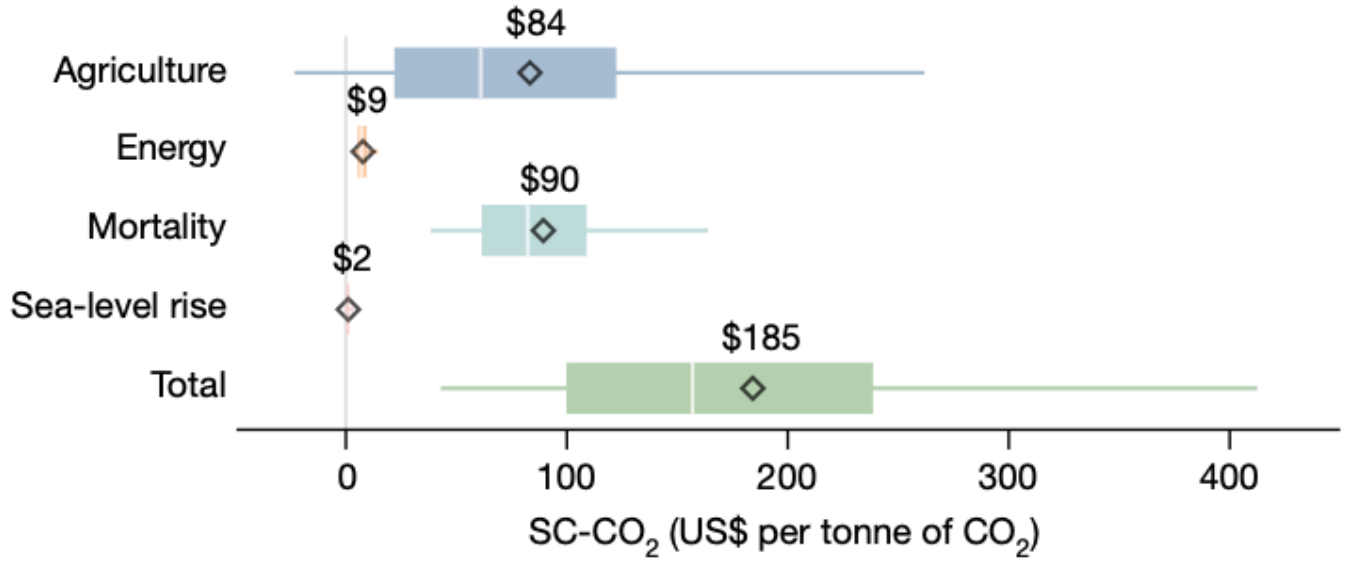


Fig. 3 | Partial SC-CO_2 estimates and uncertainty levels strongly differ across the four climate damage sectors. Box and whisker plots for the climate damage sectors included in the GIVE model, based on partial SC-CO_2 estimates for each sector. The figure depicts the median (centre white line), 25%–75% quantile range (box width), and 5%–95% quantile range (coloured horizontal lines) partial SC-CO_2 values. Black diamonds highlight each sector's mean partial SC-CO_2, with the numeric value written directly above. All SC-CO_2 values are expressed in 2020 US dollars per metric tonne of CO_2.

Takeaways on the SCC

- 2009–2017: **$40-50/ton (3% discount rate)**
- 2017–2020: **$1/ton (7% discount rate, US damages only)**
- 2021: **$51/ton (3% discount rate)**
- 2023: **$190/ton (2% discount rate)**
- 2025–today: **no federal SCC (climate damages not monetized)**
  - Discount rate matters a lot
  - Global vs. domestic damages also matters
  - Many damages are unaccounted for in current framework.
- The SCC is the single most important number for valuing the benefits of reducing climate change

Is some number better than no number?

Exploring IAMs and the SCC

- Details on the RFF modeling effort (i.e., from the Rennert et al., 2022 paper in Nature) can be found here:
  - https://www.rff.org/publications/data-tools/social-cost-of-carbon-computing-platform-models-from-the-iwg/
  - Tweak modeling assumptions: https://www.rff.org/publications/data-tools/scc-explorer

- Richard Tol has made an interactive "choose your own adventure" web app to evaluate how the SCC changes as parameter inputs change.
  - https://richardtol.shinyapps.io/SIAM3/
  - *Note: need to convert \$/tC to \$/tCO_2 by multiplying by 3.67 (also, unclear what base year for \$ is being used).*

For next class…

- Wednesday, 10/14:
  - Exam 2 review. Bring questions.
- Problem set 2 due Friday, 10/16, 11:59 PM.
- Monday, 10/19:
  - Exam 2 (classes 9–13).
  - Same format as Exam 1

- www.caseyjwichman.com/econ4210#week8